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Unlocking Abu Dhabi's World-Class Unconventional Energy Resources in Safe, Sustainable and Most Economical Way: Journey of Well Duration Reduction For Unconventional Oil Wells by 66%

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Abstract

Abu Dhabi's unconventional oil resources present a transformative opportunity for energy diversification and long-term sustainability. This paper presents Field Operator's strategic journey in reducing Oil Unconventional Appraisal PAD wells well delivery duration - There is a substantial reduction in well delivery time, with wells now being completed at 55% reduced durations, the fastest recorded at 66% reduced duration compared to previous durations against the well completed in June 2024—achieved through a combination of advanced drilling technologies, real-time data analytics, and integrated operational workflows. The initiative prioritized safety, environmental stewardship, and cost and operational efficiency, aligning with Field Operators broader sustainability goals. Key enablers included optimized well design, performance-focused execution, improved drilling practices and cross-functional collaboration. The results demonstrate a scalable model for accelerating unconventional development while maintaining operational excellence and minimizing environmental impact.

Moreover, to ensure the most effective and fit-for-purpose strategies, Field Operator has conducted comprehensive reviews of practices and performance benchmarks from unconventional giants around the globe. These insights have informed the design and execution of Company's unconventional wells, enabling the adoption of proven technologies and workflows while tailoring them to the unique geological and operational context of Abu Dhabi.

Collaborative efforts with industry partners, including Turnwell, have complemented Field Operator's internal initiatives by providing technical support and facilitating knowledge exchange, contributing to the overall success of the unconventional program

Introduction

Abu Dhabi's unconventional oil resources represent a significant frontier in the region's energy landscape. These reservoirs, while technically challenging, offer substantial potential for long-term production and

strategic energy security. Field Operator has embarked on a transformative journey to unlock these resources in a manner that is safe, sustainable, and economically viable. Historically, unconventional well delivery in Abu Dhabi faced extended durations due to complex subsurface conditions, limited benchmarking data, and evolving operational practices. Recognizing the need for efficiency, Field Operator initiated a comprehensive optimization program aimed at reducing well duration without compromising safety or environmental standards.

This manuscript outlines the strategic approach, technological enablers, and operational innovations that led to wells now being completed at 55% reduced durations, the fastest recorded at 66% reduced duration compared to previous durations against the well completed in June 2024. It highlights the integration of advanced drilling technologies, data-driven decision-making, and collaborative execution models that have redefined unconventional development in the region.

BASELINE ASSESTMENT

As baseline assessment, the historical drilling data for unconventional oil wells were analysed to establish baseline performance metrics. Most of wells were of Exploration nature with few being appraisal type with direct horizontalization which were fitting apple to apple comparison with planned PAD Appraisal wells with 6000 ft lateral length.

One of such wells were drilled in 46 days in the beginning of campaign and there was initiation to perform Workshop in reducing well durations for such wells to 30 days. This was initial milestone on well duration reduction for Unconventional Oil wells where the following main points were tackled:

- a. Section Durations for wells, compare Best of Best Value against Average
- b. Average ROP per section
- c. On bottom ROP per section
- d. W2W (Weight to Weight) Connection time
- e. Pre connection time
- f. Drilling Connection Time (S-S)
- g. Post connection time
- h. Tripping in/out speeds
- i. Tripping Connection time
- j. Casing & Frac string Running speeds (With Triple Automated catwalk in triples/doubles and Conventionally)
- k. Historical wells ILT (Invisible loss time) and challenges per Section
- l. Diverter/BOP requirements per section
- m. Reduction of flat time
- n. BHA Dynamics, Vibration levels while drilling final production wellbore in drilled wells
- o. Historical wells main NPT (Non productive Time) contributors
- p. Requirements for New Technology Deployment
- q. Operational Optimization - Streamlined rig-up/rig-down procedures
- r. Assessment of Existing casing grades and any need for downgrading and cost optimization
- s. Cluster Rigs Skidding & Rig move Optimization
- t. Well Architecture, casing setting depths
- u. Shallow loss zones, air drilling requirements, water flow from Aquifers

Additionally, Field Operator conducted a comprehensive benchmarking study by analysing unconventional drilling practices from leading operators in the United States and Saudi Arabia. The objective was to evaluate the typical well durations for unconventional wells of similar geological and

operational nature, providing a reference framework to guide performance improvement targets in Abu Dhabi's unconventional fields.

After achieving the initial 30-day well delivery target, the next milestone was set at 15 days per well. Remarkably, the fastest well duration achieved so far to date was 14.3 days—accomplished within just one year—demonstrating the rapid evolution and effectiveness of the optimization strategy. Continuous efforts are underway to further reduce well delivery durations, with the ambitious goal of achieving single-digit days per well—setting a new benchmark in unconventional drilling efficiency

Well Design Optimization

To achieve substantial reductions in well duration and improve overall performance, company implemented a rigorous well design optimization strategy tailored to the unique characteristics of Abu Dhabi's unconventional reservoirs.

Key elements included:

Amendments to Well Architecture: Field Operator introduced strategic changes to its well architecture to overcome geological challenges and improve drilling efficiency. Historically, during the early stages of the unconventional campaign, the former well architecture posed significant operational difficulties. The 9 $\frac{5}{8}$ " casing was set deep and deviated within the formation just above the unconventional reservoir target zone. This design forced drilling a 12 $\frac{1}{4}$ " hole section with a water-based mud system through a complex interval that included weak formations, deep saline aquifers, and unstable shales. Running the 9 $\frac{5}{8}$ " casing to bottom was particularly challenging, with several cases of casing getting stuck due to instability and narrow margins.

In 12.25" wellbore section, the deviation was necessary because the vertical depth available to land the well was limited. However, drilling a deviated 12 $\frac{1}{4}$ " hole section through shales introduced further complications. Increasing mud weight to stabilize the shales often led to losses in the shallower weak zones, in some cases section was drilled with caved shales.

To address these issues, there was revision on the casing strategy by setting the 9 $\frac{5}{8}$ " casing shallower—above the shales—in a competent tight limestone cap rock. This adjustment allowed the casing to cover the shallower, weaker aquifers and eliminated the need to drill the problematic section with water-based mud. It also enabled the use of oil-based mud systems in the subsequent production hole section as with former architecture Oil based system was not preferred in same hole section with Aquifers.

With this new architecture, the curve and lateral sections of the well were drilled in the 8 $\frac{1}{2}$ " production hole using an oil-based mud system. This change significantly improved wellbore stability, enhanced rate of penetration (ROP), and reduced non-productive time (NPT), contributing directly to the reduction in overall well duration.

Additionally, setting 9.5/8" casing shallower ended up with 2500 ft of tangibles cost saving

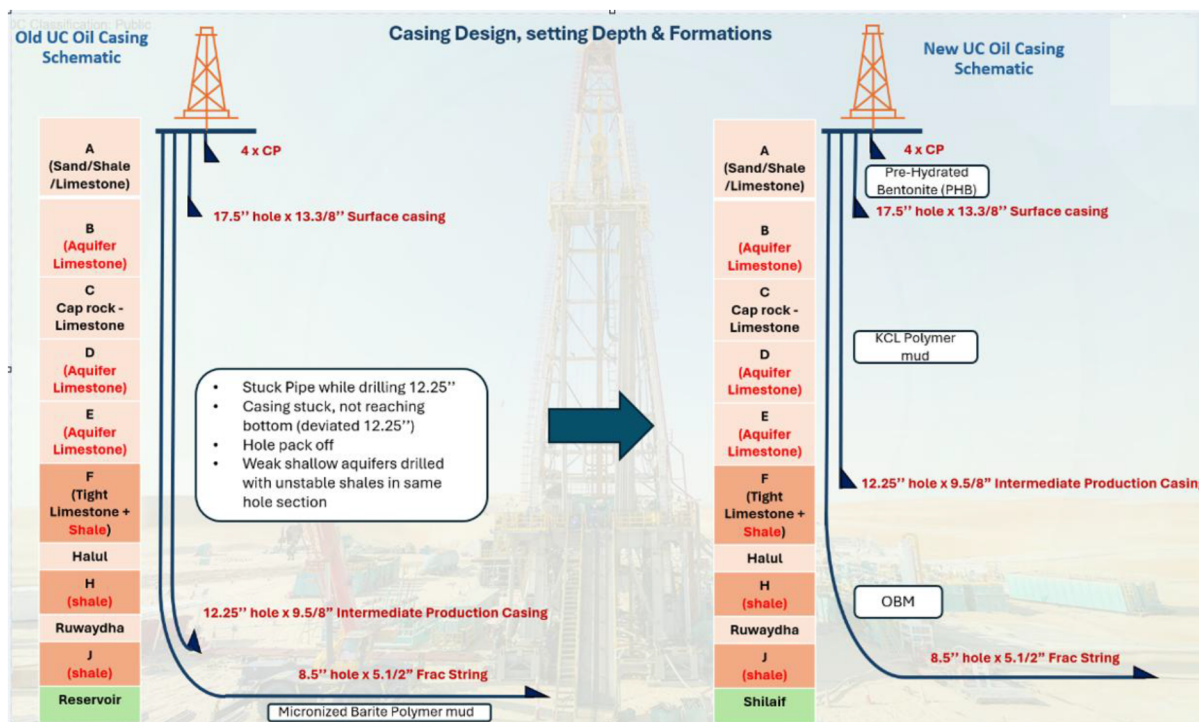


Figure 1—Display of old and new well Architecture for Unconventional Oil well

Casing Design & Wellhead

Casing strings were engineered to balance mechanical integrity with cost efficiency. Company still under study of downgrading casing strings more down to make it fit for purpose and economical as Appraisal stage phase 1 is being completed

There are more intentions of downgrading below Casing strings in Development phase plus with updated required surface fracturing pressures post appraisal/exploration wells:

- 13.3/8", L80, 68 PPF, Premium down to 13.3/8", J-55, BTC (already completed)
- 9.5/8", L80, 47 ppf, Premium down to 9.5/8", L80, 43.5 PPF, Premium or Butress (In progress)
- 5.5" Frac string, 26 ppf, T95, Premium down to 5.5", 23 ppf, P-110 mixed with T95 (In progress)

Trajectory Modifications

Company revised its trajectory designs to achieve more efficient drilling paths and improve overall well performance. These changes were driven by both geological constraints and strategic land utilization goals.

Initially, under the former well architecture, the 9 $\frac{5}{8}$ " casing was run deviated at inclinations of around 50–60 degrees, landing just above the unconventional reservoir. The deviation at such depths, combined with unstable formations, made it difficult to run casing to bottom, with several instances of casing getting stuck.

As Company shifted its strategy to maximize the utilization of unconventional acreage and avoid losing valuable land, the vertical sections for landing points were reduced by nearly 50%. This required a more aggressive trajectory design with tighter build sections and reduced vertical depth. There was introduction of RSS (Rotary steerable system) with higher DLS (Dogleg severity) capability and shanckless bit which resulted in higher DLS capabilities than previously used RSS + bit combination, to cater for shortened landing point vertical sections therefore forcing for negative displacement, additionally some formations in curved section have low UCS leading lower maximum DLS achievable by RSS. It was observed also that the negative displacement has positive effect on frack string deployment to well TD.

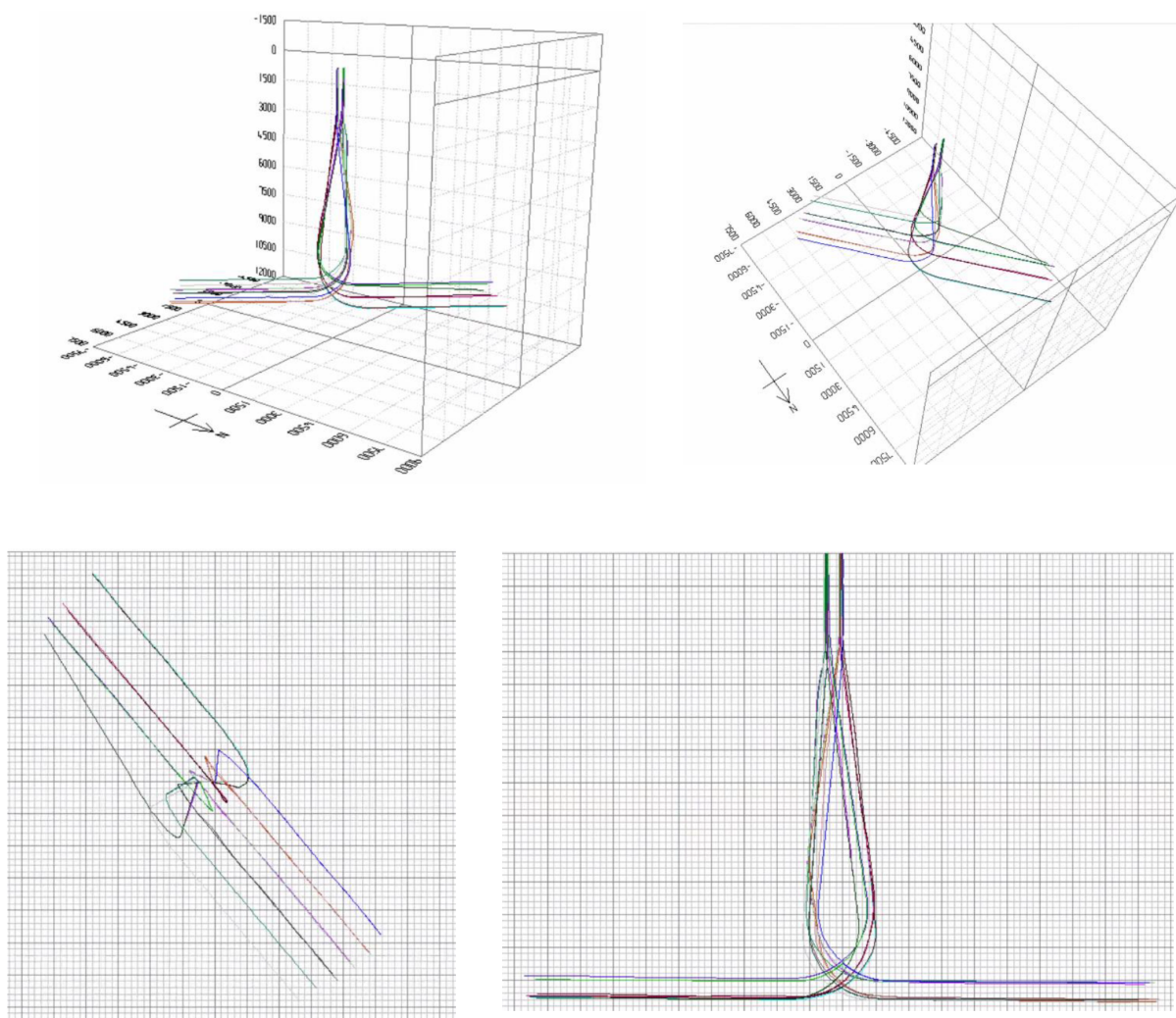


Figure 2—Sample of UC OIL PAD trajectory

There was introduction of new RSS push the bit system with short make up drill bit which improved DLS in curves, efficient turning of steering energy into build up rates.

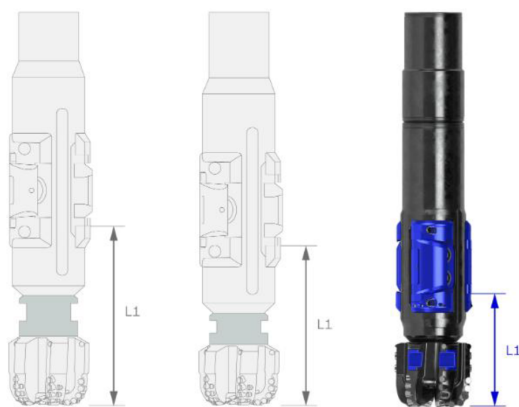


Figure 3—The short make up bit for sharper turns compared to conventional bits

Mud System & Density Optimization

A major shift in the production hole section involved transitioning from a water-based mud system to an oil-based mud system. This change improved wellbore stability, enhanced ROP, and reduced NPT, especially in challenging formations.

One main challenge was to get familiar with unconventional type of Reservoirs and required overbalance to drill. As the Unconventional reservoir and campaign was new, the conventional wells would require minimum 300 psi overbalance as per company policy. Initially wells were being drilled following 300 psi Overbalance policy with oil based mud system and high Mud weights, Reduced ROP, higher solids content in mud leading to MWD tools jamming, mud pump liners leaking, higher pump pressures and several other problems.

As Appraisal well were being drilled in PAD in batch drilling mode, there was Mud density reduction from first well to last well on same pad in gradual way and mud density was reduced by 20 % which lead to drill underbalanced from high end Pore Pressure and overbalance on low end Pore pressure as formation pressure data if well proposal was provided in range.

New mud weight was considered above the formation collapse pressure and create no wellbore stability issues across shales penetrated in production hole in curve section

It is worth to mention, with any mud weight, there are still some operational issues faced at shale to Limestone and Limestone to shale transitions which are under study and assumed to be hole ledging issues in those transitions

Cluster Rigs Skidding& Rig move Optimization

Change in Skidding strategy by shifting from single to double section drilling – Reducing # of skids per PAD (5 wells) from 12 to 8 skids, which helped in reducing number of skids per PAD and cost saving. Formerly, batch drilling technique was to drill section by section, later combined top 2 section and started drilling wells in double sections.

This was performed in a way that, handling mud system does not impact operations. Also same mud system can be used for upcoming wells on same PAD which enhanced sustainability as well.

- **Completion Strategy Alignment:** The well design was closely integrated with the planned completion strategy, ensuring that stimulation requirements (e.g., hydraulic fracturing stages) were supported by the well architecture. This alignment minimized rework and improved post-drilling efficiency.
- **Geomechanical Modelling and Offset Analysis:** Advanced geomechanical models and offset well data were used to predict formation behaviour, optimize casing points, and mitigate drilling risks such as wellbore instability and differential sticking.

Drilling Performance Enhancement

Company implemented a series of targeted drilling performance enhancements to reduce well durations and improve operational efficiency across its unconventional campaign. These improvements were driven by lessons learned from early wells, benchmarking global best practices, and continuous field-level optimization.

Key initiatives included:

Bit, BHA and ROP Optimization: To simultaneously enhance wellbore stability and rate of penetration (ROP), the optimization of the bottom hole assembly (BHA) design was strategically paired with bit design modifications. This integrated approach allowed for improved directional control, minimized shocks, vibration and stick-slip tendencies, and maximized drilling efficiency in unconventional formations, especially in 8.5” final production wellbore. Additionally there was utilization of Intelligent driller

advisory and Autonomous directional drilling on some wells, which had positive effect on ROP, trajectory smoothness with lower tortuosity and frac string deployment.

Main performance improvement approach on bit design and optimization was as below:

- The PDC Bit design trial tested 3D cutter shapes and different blade and cutter size configurations, leading to a final design with a unique layout of 2D and 3D cutter shapes
- A cutting-edge and durable drive system designed to meet application objectives, setting new standards in drilling dynamics and performance
- Ongoing performance analysis and learning from service quality incident

The main approach for BHA design for 8.5" hole was as follows:

- Stay within intended layer during Geosteering and stability of BHA in sub layers with different UCS and GR layers.
- Improved ROP
- Achieve Required DLS in curve section, land well on same section and drill well to TD with same BHA on one run, considering high Delta Temperature from 9.8" casing shoe to Landing reservoir with approximate temperature difference of +/- 150 deg F
- Rock characterization for each layer and BHA response, proper parameter roadmap development for each formation
- Static and dynamic Temperatures, mitigations in case of high temperatures

With ROPs ranging from 40-50 ft/hr historically, with optimization, ROP observed at highest at 124.25 ft/hr



Figure 4—ROP Improvement curve

Drilling Parameter Tuning: Real-time monitoring and adjustment of drilling parameters—such as weight on bit, rotary speed, and mud flow rate—enabled proactive management of downhole conditions. This helped mitigate vibration, stick-slip, and other dysfunctions, contributing to a 15–20% reduction in mechanical downtime. The primary approach focused on increasing drilling parameters, specifically operating rigs at a higher percentage of their design limits. By pushing weight on bit (WOB), revolutions per minute (RPM), and pump rates closer to optimal thresholds, the team was able to unlock significant performance gains while maintaining safety and equipment integrity. This proactive strategy laid the groundwork for subsequent improvements in rate of penetration (ROP) and overall well delivery efficiency.

For 8.5” section, main concerns on shocks and vibrations were Axial shock peaks at MWD and Lateral Vibration at RSS push the bit system. MWD axial shock peaks were mainly controlled by introduction of suppressors and RSS lateral Vibrations were mainly controlled by bit design and adjusting parameters with proper roadmap. To sustain tool life and avoid fatigue, a proactive shock and vibration monitoring system was implemented, structured around four predefined risk levels:

Levels	Zone Risk Description	Actions
Level 1	Safe Zone	Normal operating conditions with minimal shock and vibration
Level 2	Risk Balance Zone (<i>Target Zone</i>)	Optimal drilling performance with acceptable tool stress. Operations were planned to remain within this zone.
Level 3	Elevated Risk Zone	When shock and/or vibration levels reached this threshold, mandatory mitigation actions were triggered. The primary response involved: Reducing Weight on Bit (WOB) Adjusting RPM to identify a new optimized operating window
Level 4	Critical Zone	If both shock and vibration exceeded this level Drilling to be immediately halted String to be picked up off bottom to release accumulated stress Parameters to be reset , and drilling resume gradually under controlled conditions

Figure 5—Shocks and Vibrations Limits

This tiered approach ensured real-time response to dynamic downhole conditions, preserving tool integrity while maintaining aggressive drilling performance.

In support of this, extensive studies were conducted on PDC drill bit response to geomechanical heterogeneity within the unconventional reservoir sublayers, particularly in the 8½” hole section. These studies helped optimize bit selection and drilling parameters for varying lithologies, leading to more stable drilling performance and reduced bit wear. Basically, Oil Unconventional reservoir has 2 main Layers to land, with deeper layer has 3 sub layers which has different TOC (organic matter) compressive strength and porosity. Upper Layer has only 1 sub Layer

Upper main Layer featured with Low GR with UCS ranging from 15-20Ksi, also requires higher DLS. Lower main Layer thicker layer than upper with multiple GR layers which is also subjected to sub sub layers where this application allowed maximum 50 ft TVD of Geosteering. Also additional challenge in curve section was 2 formations with low UCS which gave maximum 6 DLS with available Push the bit system.

This presented significant heterogeneity in mechanical and geochemical properties across its sublayers. Each sublayer exhibits distinct gamma ray responses, variable organic matter content, and differing unconfined compressive strengths (UCS), directly impacting drill bit behavior, rate of penetration (ROP), wear mechanisms, vibration severity and overall drilling efficiency.

Investigating the performance of various polycrystalline diamond compact (PDC) while drilling through these sublayers in multiple unconventional oil wells. By correlating real-time drilling data with

petrophysical and geomechanical logs, the research identifies bit performance trends linked to formation characteristics.

From the statistics on drilled UC oil wells, the main challenging sub sub layers were identified as below mainly due to higher UCS with higher density, high Shocks and Vibrations. Slower overall ROP due to Vibration mitigation parameters:

- Lower layer, lowest sub layer
- Lower layer, top sub layer
- Upper Layer, main sub layer

Main suggestion was to have Suppressor Vibration Dampening tool as standard BHA and new bit design. Below is main table indicating overall ROP in terms of response to GR content indicating higher GR content layers have more ROP.

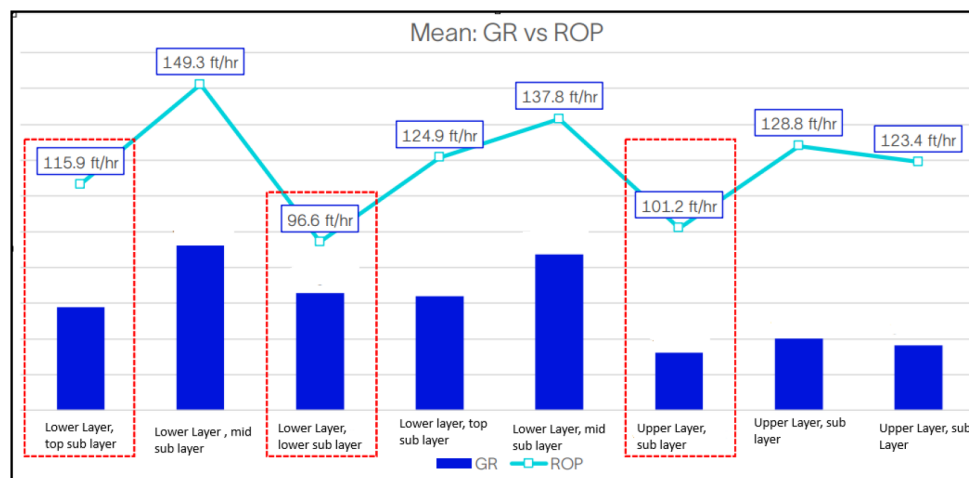


Figure 6—ROP vs GR content on sub layers

It can be also observed that, Lower layer, mid sub layer is easiest one to drill which has highest GR content, lowest UCS and lower shocks and vibrations among the sub slayer of lower layer and among all layers mainly drilled with 516 PDC bit which is more aggressive compared to 616 PDC bit. Other Sub layers with Lower GR, higher UCS required 616 PDC bit with Suppressor.

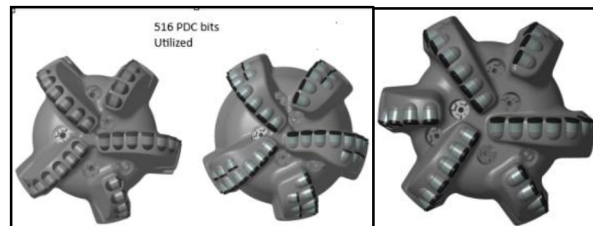


Figure 7—some of used 516 and 616 PDC bits

The findings underscore the importance of dynamic bit selection strategies based on detailed sublayer characterization. Recommendations include real-time lithology recognition integration, localized bit design customization, and adaptive drilling parameter management. By aligning bit design and drilling practices with sublayer-specific conditions, operators can enhance drilling efficiency, reduce costs, and minimize borehole instability in challenging unconventional targets within the Reservoir.

3d cutters- In unconventional reservoirs, especially where formations vary in UCS and abrasiveness, 3D cutters can significantly improve drilling efficiency and bit life, especially when paired with RSS tools

Rig and Equipment: Company deployed cluster friendly single row skidding high-spec rigs to support faster and safer operations.

To further enhance efficiency, skidding rigs were introduced to the unconventional program, enabling batch drilling from multi-well pads. This significantly reduced rig move time between wells and improved overall pad productivity.

Additionally, offline tubulars bucking services were implemented in conjunction with Triplex catwalk systems, allowing tubulars to be made up and staged offline. This not only reduced tripping time and improved safety by minimizing manual handling, but also enabled efficient drill pipe management during drill pipe and tubular inspections in pad drilling operations.

These systems are also expected to play a critical role in future campaigns, especially as well laterals are extended to maximum lengths. The combination of Triplex catwalks and bucking units will help mitigate setback capacity limitations, ensuring smooth and efficient pipe handling and storage throughout extended lateral drilling operations.

Flat Time Reduction Strategies: Company focused on minimizing non-drilling time through improved rig move procedures, faster casing running, and streamlined cementing operations. Overall flat time was reduced by 25–35%, contributing directly to shorter well durations.

A major contributor to this improvement was the implementation of batch drilling techniques from multi-well pads which gave main below benefits. This approach enabled a wide range of offline operations, but not limited to:

Item	Description
Offline wellhead and X-tree make-up	Reducing rig-time dependency
Offline cement evaluation logging in the 9 ⁵ / ₈ " casing	Well duration Reduction, allowing early assessment of cement integrity without interrupting drilling schedules
Improved crew experience and efficiency	Due to repeated execution of similar operations across wells on the same pad
Enhanced logistics coordination and mud system repeatability	Which streamlined material delivery and fluid management across multiple wells
Use same BHA for different wells	In some cases when single section batch drilling was implemented, top hole BHA was racked back and utilized on next well
Extension of BOP pressure test frequency from 14 days to 21 days for the 12 ¹ / ₄ " hole section. Company adopted API BOP test requirements tailored for pad drilling operations	BOP stack is fully tested on the first well of the pad, and only connection tests are performed on subsequent wells—provided the tests remain valid within the 21-day window. This approach significantly reduced flat time associated with repetitive BOP testing while maintaining safety and compliance.
Elimination of Diverter Utilization in surface hole	Earlier wells utilized Diverter in top hole due to presence of shallow fresh water and some aquifers. After analysing offset wells, 13.3/8" casing post cementing behaviours and any shallow gas anomalies the diverter utilization from surface hole was removed saving rig time and cost
Offline cementing of 13.3/8" or 9.5/8" casing	Studies are still ongoing

Figure 8—Flat time reduction

In addition, significant improvements were made in skidding operations. Initially, skidding rigs from well to well on the same pad was performed only during daylight hours and often took extended durations. To eliminate delays caused by waiting for daylight, night skidding was introduced using the HALO light system, enabling safe and efficient nighttime rig moves.

Furthermore, offline skid preparations—such as disconnection of the catwalk and pre-installation of skidding beams while drilling the current well—were implemented. These changes dramatically reduced skidding durations from 36 hours to just 5 hours, contributing to faster transitions between wells and improved overall pad efficiency.

To streamline rigless intervention operations and reduce non-productive time, a novel approach was implemented during frac string cementing. Instead of using conventional mud with solids, fresh water was used to displace the frac string cement, significantly simplifying the displacement process and reducing cleanup requirements.

Due to the increased horsepower demand of this method, two cement units were deployed:

- One primary unit for cementing operations
- A mobile backup unit ("flying unit") assigned to circulate among rigs within the same field

Apart from benefits of batch drilling offline operations, below are more well duration reduction strategies implemented:

b) W2W connection time analysis and reduction: Close follow up on weight to weight connections by analysing pre connection time, drilling connection time and post connection time lead to tremendous reduction in flat time reduction. Introduction of battery based MWD system with built in shock sub enhanced it further with no survey time in connections.

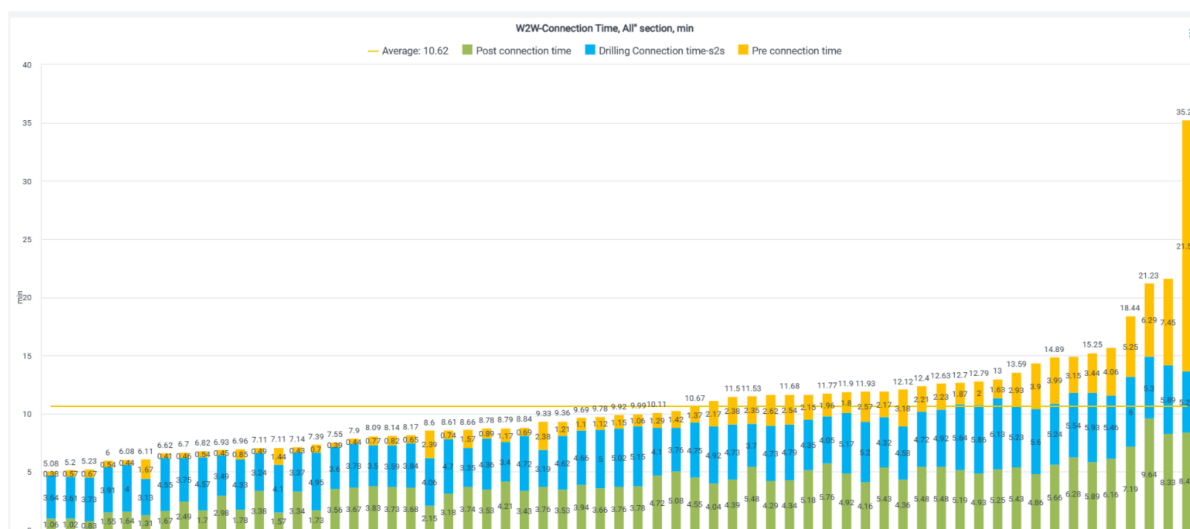


Figure 9—Improvement of W2W connection time over a period

c) Connection practices – Historically, full washing up and reaming down was practiced after each prior each connect which increase W2W durations drastically. Later on this practice was halted with proper connection practices depending hole and angle and pressure & drag conditions faced after drilling down stand. If high torque and pressure observed after stand down, only half single backreaming was performed which reduced pre-connection duration drastically. Post connection time was also drastically reduced with optimization of surveying procedure. No waiting for complete frame from MWD, once synced, RPM was started and BHA was run to bottom and resumed drilling. In case of battery based MWD, there was no requirement to wait for MWD sync, once pump pressure stabilized, drilling to be started.

For vertical or near vertical sections, normal survey procedures were followed, in case of any survey missed, surveys to be taken on next stand and waited for complete survey frame. But in build or curved sections valid survey to be taken on each stand. If survey missed in build or curve sections, surveys were to be re-taken again. When it came to lateral section, after landing the well, normal survey procedures applied but if survey missed, survey to be taken on next connection with complete survey frame.

d) Reduction of backreaming across Aquifers in 12.25" hole section: There was substantial study performed on reduction of backreaming across shallower Aquifers to avoid excessive backreaming. Below are some trials performed with improvement and studies still ongoing:

- Utilization of drilling reamer tools (Uni & Bi direction), Roller reamers, Dog leg reamers in different combinations with stabilizers
- Utilization of Eccentric reamers
- Drilling with different mud systems such as KPG, Low vis, brine + KPG, Brine, 3% KCL Brine + KPG

Drilling Fluid Management: The transition to oil-based mud systems in the production hole section significantly improved wellbore stability and ROP. Compared to earlier wells drilled with water-based mud, the oil-based system reduced non-productive time (NPT) by 40–50% and enabled more consistent drilling performance through unstable formations.

Remote Operations and Real-Time Support: Company leveraged remote operations centres to monitor drilling activities in real time, enabling faster troubleshooting and performance optimization. This approach contributed to a 10–15% improvement in decision-making speed and reduced response time to downhole challenges.

Operational Efficiency Measures

In addition to technical and rig-based improvements, Company implemented a series of operational efficiency measures that played a critical role in reducing well durations and enhancing overall campaign performance. These measures focused on optimizing logistics, workflows, and cross-functional coordination.

Key initiatives included:

Streamlined Logistics and Material Management: Improved coordination between supply chain teams, service providers, and rig crews ensured timely delivery of critical materials such as casing, cement, and drilling fluids. Pre-staging of equipment and proactive inventory management reduced waiting times and minimized disruptions. These efforts contributed to a 15–20% reduction in logistics-related delays.

Integrated Planning and Scheduling: Cross-functional planning sessions aligned drilling, completion, and logistics teams, allowing for dynamic scheduling adjustments and early identification of bottlenecks. This led to a 10–15% improvement in overall operational flow across multi-well pads.

Crew Training and Standardization: Repeated operations across batch-drilled wells enabled crews to gain familiarity and efficiency. Standard operating procedures (SOPs) were refined and reinforced through targeted training, resulting in faster execution and a 20–25% reduction in procedural errors.

Digital Dashboards and KPI Tracking: Real-time dashboards monitored key performance indicators (KPIs) such as ROP, flat time, NPT, and rig move durations. This enabled data-driven decision-making and rapid response to emerging issues. KPI visibility helped reduce troubleshooting time by up to 30%.

Pre-Job Planning and Lessons Learned Integration: Detailed pre-job planning sessions were conducted for each well, incorporating lessons learned from previous wells on the pad. This ensured continuous improvement and reduced the likelihood of repeating past inefficiencies. Implementation of these learnings led to consistent cycle time improvements across pads.

Enhanced Collaboration with Service Providers: Company worked closely with service companies to align expectations, improve accountability, and ensure readiness for each phase of the well. This collaboration helped reduce service-related delays and improved execution quality, contributing to a 10–20% improvement in service delivery timelines.

Results

The implementation of strategic design changes, drilling performance enhancements, and operational efficiency measures led to substantial improvements in Field Operator's unconventional drilling campaign. These results reflect the effectiveness of a data-driven, collaborative, and continuously evolving approach.

Key outcomes include:

- **Well Duration Reduction:** The average well duration was reduced by up to 66% compared to early campaign wells, marking a significant milestone in Field Operator's unconventional journey.
- **Improved Rate of Penetration (ROP):** Optimized bit and BHA designs, combined with oil-based mud systems and tuned drilling parameters, resulted in 35–50% higher ROP in the 8½" production hole section.
- **Reduced Non-Productive Time (NPT):** Transitioning to oil-based mud and improving wellbore stability led to a 40–50% reduction in NPT, especially in shale and aquifer zones.
- **Skidding Time Optimization:** Skidding durations were dramatically reduced from 36 hours to just 5 hours through night skidding and offline preparations, enabling faster transitions between wells on the same pad.
- **Flat Time Reduction:** Batch drilling and offline operations contributed to a 25–35% reduction in flat time, streamlining well delivery and improving pad efficiency.
- **Operational Consistency:** Standardized workflows, crew training, and digital KPI tracking enabled consistent performance across pads, reducing variability and improving predictability.
- **Enhanced Safety and Compliance:** Adoption of API BOP testing protocols and improved rig systems ensured that efficiency gains were achieved without compromising safety or regulatory compliance.

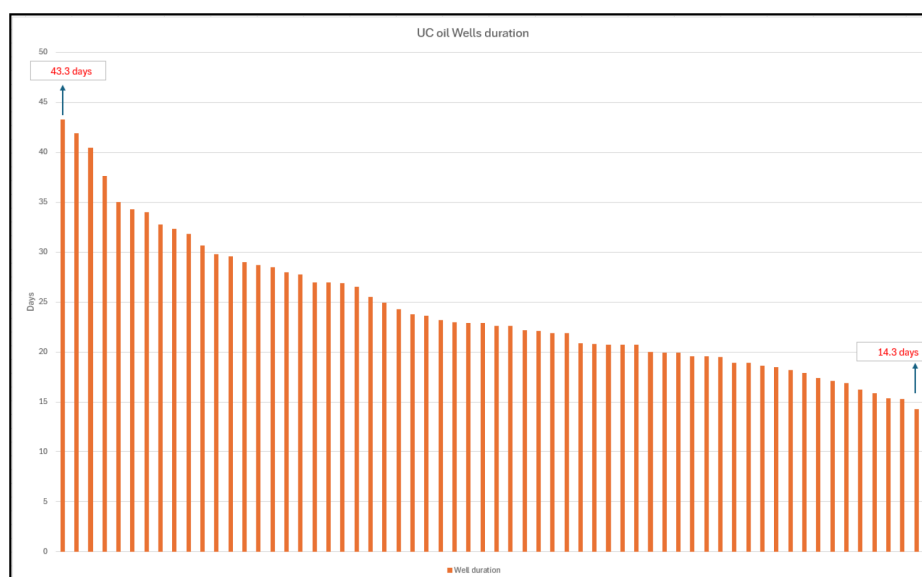


Figure 10—Wells duration Reduction chart

These results not only demonstrate Field Operator's ability to adapt and innovate but also position the organization to scale its unconventional operations sustainably and economically in future campaigns.

Conclusion

Field Operator's unconventional drilling campaign has undergone a transformative journey marked by strategic innovation, operational discipline, and continuous learning. By revisiting and optimizing well architecture, refining trajectory designs, enhancing drilling performance, and streamlining operational workflows, Field Operator successfully reduced well durations by up to 66%, setting a new benchmark for efficiency in unconventional development.

The integration of global best practices, field-specific learnings, and collaborative partnerships has enabled Field Operator to overcome complex subsurface challenges while maintaining safety, sustainability, and cost-effectiveness. The adoption of batch drilling, night skidding, oil-based mud systems, and advanced rig technologies has not only improved current performance but also laid a strong foundation for future campaigns, especially as lateral lengths and well complexity continue to increase.

These achievements reflect Field Operator's commitment to unlocking Abu Dhabi's world-class unconventional energy resources in the safest, most sustainable, and economical way. The lessons learned and systems established through this journey will continue to drive performance improvements and support Company's long-term vision for unconventional resource development.

Abbreviations and acronyms

BHA	Bottom Hole Assembly
KPI	Key Performance Indicator
ILT	Invisible lost time
NPT	Non-Productive Time
ROP	Rate of Penetration
UC	Unconventional
W2W	Weight to Weight
BOP	Blowout preventer
PPF	Pound per foot
BTC	Buttress Thread Casing
RSS	Rotary Steerable System
RPM	Rotations per minute
MWD	Measurement while drilling
PDC	Polycrystalline diamond compact
TOC	Total organic carbon
GR	Gamma Ray
API	American Petroleum Institute
NPT	Non Productive Time
TD	Total depth

References

- Field Operator's Onshore Drilling Manual - Vol 1 - Chapter-1 Policies - 2018 (3)
- Field Operator's, Drilling Function Unconventional (UC) *Drilling & Frac Manual* Volume – 3
- API RP 53 A 01 MAR 1997 Blowout Prevention Equipment Systems for Drilling Wells Third Edition <https://jpt.spe.org/>-
The new award accompanies the creation of an unconventional drilling subsidiary and potential equity deals with SLB and Patterson-UTI.
- Field Operator's - Well Integrity Management Standards (WIMS)
- Rakesh Roshan Rana, (November 04 2024) Optimizing Exploitation Strategies for UAE Unconventional Oil Resources: A Case Study on Integrating Rock Quality & Completion Quality Aspects, SPE-222910-MS, <https://doi.org/10.2118/222910-MS>
- Nabila Lazreq, (Society of Petroleum Engineers - Abu Dhabi International Petroleum Exhibition and Conference 2020, ADIP 2020), Tight oil from shale rock in UAE: A success story of unconventional fracturing,